ModelIt!

Student Activity Pack

Why Can't Two Plants Share the Same Sunlight?

Modeling Competition for Limited Resources in Ecosystems

Grade 06L2 | NGSS MS-LS2-1, MS-LS2-2 | Cause and Effect

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **If you plant two different types of plants very close together, what do you think will happen over time?**

2. **Why do you think some plants grow tall while others stay close to the ground?**

3. **What resources do plants need to survive and where do those resources come from?**

4. **Draw what you think happens to sunlight as it passes through a forest with tall and short plants.**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Sunlight Available | Soil Nutrients | Plant A Growth | Plant B Growth | Resource Depletion

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Equal Start

Both plants at equal size, moderate resources

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: One Plant Advantage

Plant A starts taller, same soil nutrients

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Nutrient Crash

Lock Soil Nutrients to low

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Water Availability — The amount of water in the soil, which both plants need and which can become a limiting factor in dry conditions

Root Depth Difference — How deeply each plant's roots extend into the soil. Different root depths can allow plants to access different nutrient zones

Allelopathy — Chemical compounds released by some plants that inhibit the growth of nearby competitors, a form of chemical competition

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

If there is plenty of sunlight hitting a forest, why can't all plants grow equally tall and healthy?
Use evidence from your simulation to support your answers.

1. **Why does even a small head start in height give one plant a compounding advantage over the other?**

2. **How is competition between plants different from competition between animals?**

3. **What would happen in your model if both plants were identical in every way?**

4. **How does resource partitioning allow species that would otherwise compete to coexist?**

5. **Why does competitive exclusion not eliminate all biodiversity in natural ecosystems?**

8 STEM Challenge: Design a Coexistence Garden

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YOUR ENGINEERING MISSION

Design a small garden where two competing plant species can coexist by using resource partitioning strategies.

THE SCENARIO

A community garden has limited space and wants to maximize biodiversity by growing two plant species in the same bed. Your ecology team must design a planting arrangement that allows both species to thrive by minimizing direct competition for sunlight and nutrients.

Engineering Design Thinking

1. WHAT TWO PLANT SPECIES WILL YOU CHOOSE AND WHAT ARE THEIR LIGHT AND NUTRIENT REQUIREMENTS?

2. HOW WILL PLANTING ARRANGEMENT (SPACING, HEIGHT TIERS, ORIENTATION) REDUCE LIGHT COMPETITION?

3. WHAT SOIL AMENDMENTS OR ROOT ZONE STRATEGIES WILL HELP BOTH PLANTS ACCESS NUTRIENTS?

4. HOW WILL YOU MONITOR THE HEALTH AND GROWTH OF BOTH SPECIES OVER SEVERAL WEEKS?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

Explain your design using scientific evidence from the simulation:

Real Career Connection

Ecologists and Conservation Biologists study species interactions including competition to protect biodiversity and manage ecosystems. They earn $55,000-$110,000/year.